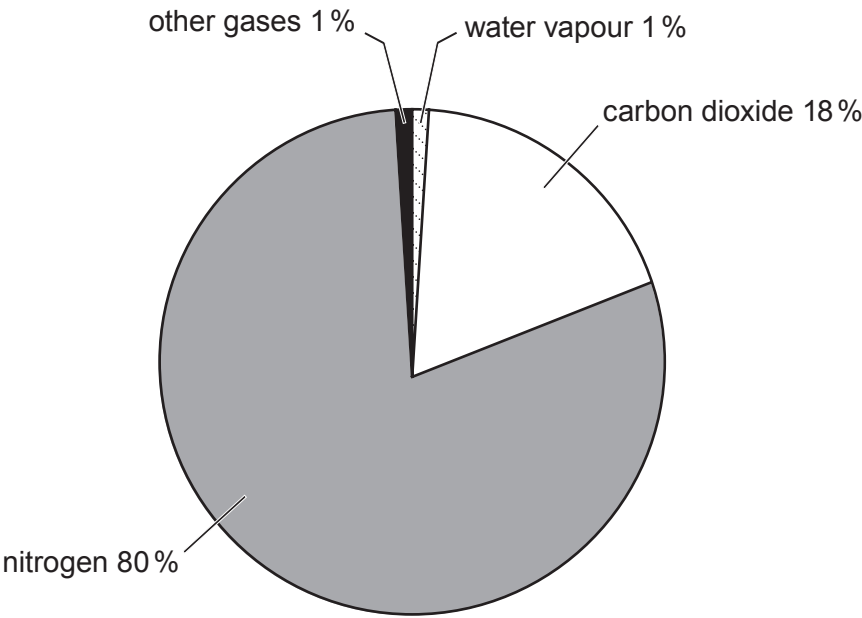


7. (a) The Earth’s atmosphere has evolved over millions of years to its present composition. The pie chart shows the composition of the atmosphere at one stage of its evolution.



- (i) Describe the **main** differences in the composition of the atmosphere shown in the pie chart and that present on the Earth today. [3]

.....

.....

.....

.....

- (ii) **Complete the table** to describe simple chemical tests that can be used in the laboratory to identify hydrogen and carbon dioxide. [2]

Gas	Test carried out	Expected observation
hydrogen		
carbon dioxide		



Examiner
only

- (b) During the last 250 years the level of carbon dioxide in the atmosphere has increased. Most scientists believe that the increase in the level of carbon dioxide in the atmosphere has resulted in global warming.

The table below shows some data about the atmosphere between 1750 and 2000.

Year	Concentration of carbon dioxide in the atmosphere (parts per million)	Average global temperature (°C)
1750	278	13.3
1800	282	13.4
1850	288	13.5
1900	297	13.7
1950	310	14.0
2000	368	14.6

Use the data in the table to answer parts (i) and (ii).

- (i) Describe the trend in average global temperature between 1750 and 2000. [2]

.....

.....

.....

- (ii) Calculate the percentage increase in the concentration of carbon dioxide in the atmosphere from 1750 to 1850. [2]

percentage increase = %

9



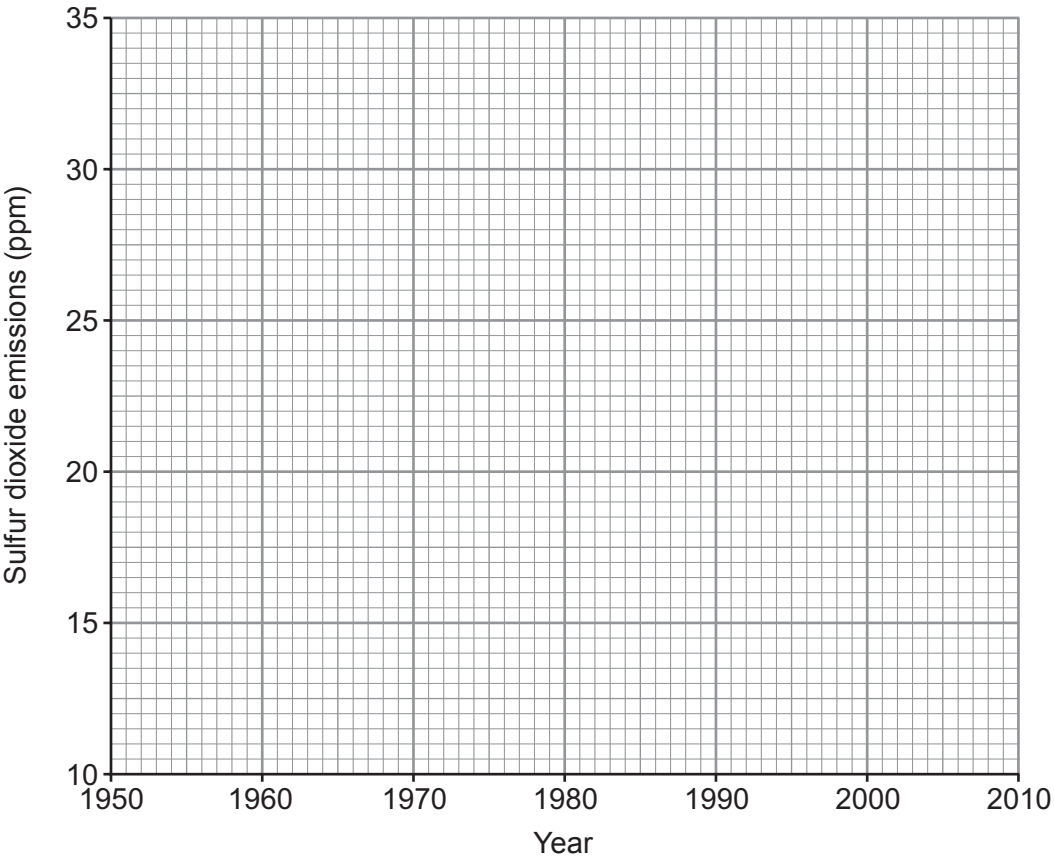
7. Burning fossil fuels containing sulfur causes sulfur dioxide, SO₂, to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

Year	Sulfur dioxide emissions (ppm)
1950	12.0
1960	16.0
1970	21.5
1980	29.5
1990	29.0
2000	24.0
2010	18.5

ppm = parts per million

(a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]



- (ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

.....

.....

.....

- (iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

.....

.....

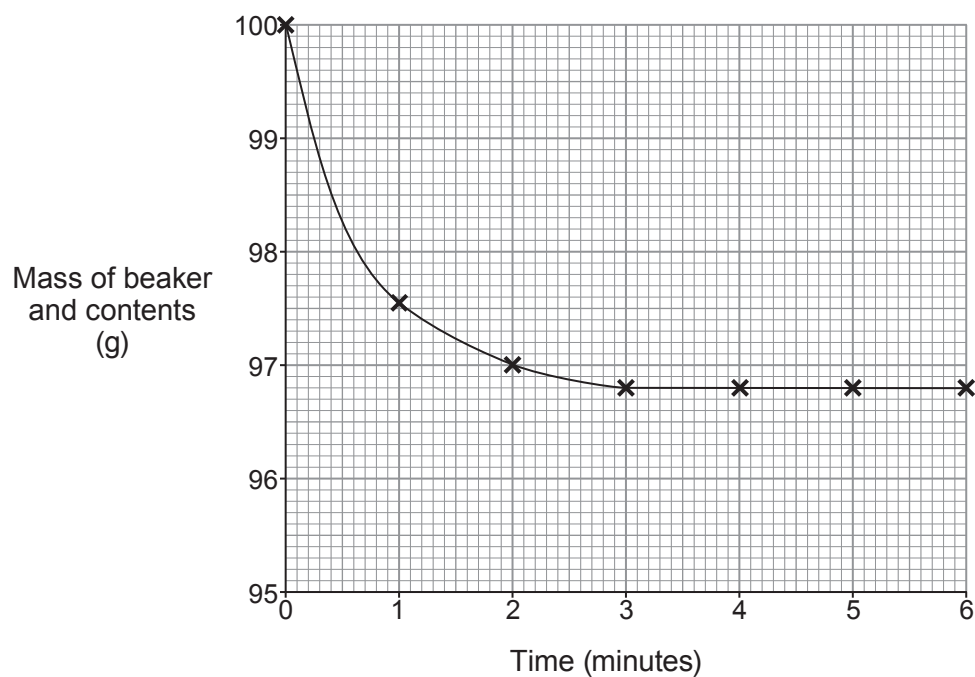
.....

- (b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H_2S .

Complete and balance the symbol equation for this reaction. [2]



7. Powdered calcium carbonate was added to an excess of dilute hydrochloric acid in a beaker. The mass of the beaker and its contents was recorded every minute for 6 minutes. The graph shows the results.



The equation for the reaction is



- (a) Describe how the mass of the beaker and its contents changes over the first minute. Give the reason for this change. [2]

.....

.....



- (b) (i) Use the graph to determine the time taken for the reaction to finish. [1]

- (ii) When is the reaction at its fastest? Tick (✓) the correct box. [1]

from 0 – 0.5 minutes

☐

from 1 – 1.5 minutes

☐

from 2 – 2.5 minutes

☐

from 3 – 3.5 minutes

☐

- (c) Use your graph to calculate the mean rate of the reaction during the **first two minutes**.
Use the following equation. [2]

$$\text{rate} = \frac{\text{decrease in mass of beaker and contents (g)}}{\text{time (minutes)}}$$

Rate = g/minute

- (d) The experiment was repeated using the same mass of calcium carbonate but as a **lump** instead of a powder.

On the grid opposite, sketch the graph you would expect to obtain from this second experiment. [1]

7



Examiner
only

7. Describe the advantages and disadvantages of hard water. [6 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

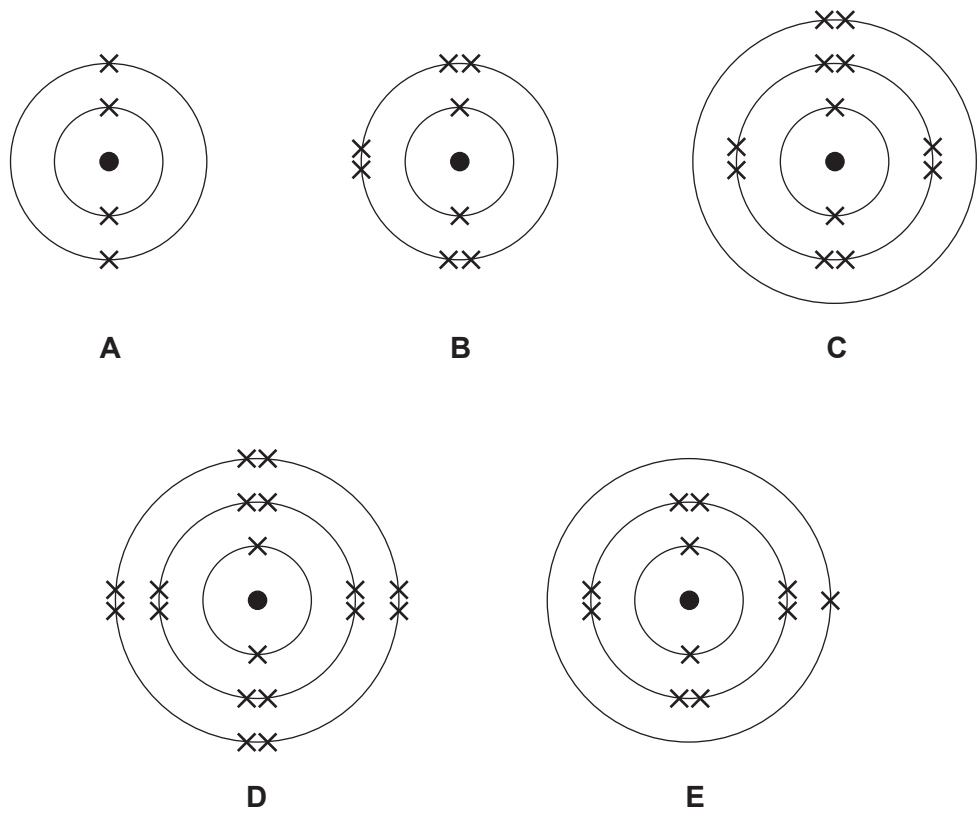
.....

.....

6



7. The diagrams below show the electronic structures of five elements, **A**, **B**, **C**, **D** and **E**.
The letters are not the symbols of the elements.



- (a) Give the **letters** of the elements found in Period 2 of the Periodic Table. Give a reason for your choice. [2]

Letters and

Reason

.....



- (b) Give the **letter** of the element found in Group 0 of the Periodic Table. Give a reason for your choice. [2]

Letter

Reason

- (c) Explain how the electronic structure of element **E** can be used to determine its atomic number. [2]

.....
.....
.....

- (d) One of the diagrams represents the element oxygen. Oxygen reacts with potassium to form potassium oxide.

Give the formula for potassium oxide and balance the equation for this reaction. [2]

